Transport Phenomena, R. Byron Bird, Warren E. Stewart, Edwin N. Lightfoot (2nd Edition):

Chapter 11: 11A.1, 11.B2, 11B.4

Chapter 12: 12.B.4

A solid sphere of radius R is initially at a uniform temperature T_0. At time $t = 0$, the temperature of the sphere's surface is suddenly raised to T_1. Assume the density, heat capacity, and thermal conductivity of the solid are constant.

1- Write down the partial differential equation governing the transient temperature profile within the sphere.

2- Rewrite the differential equation in terms of dimensionless variables: $\Theta = \frac{T_1 - T}{T_1 - T_0}$, $\eta = \frac{r}{R}$, and $\tau = \frac{\alpha t}{R^2}$.

3- Solve the dimensionless partial differential equation using the method of separation of variables. Express the dimensionless temperature profile as an infinite series and evaluate the integration constants.

4- Plot the dimensionless temperature as function of dimensionless radius for $\tau = 0.1$.

5- Modify the provided finite difference code for the spherical coordinates and reproduce the above plot.

11.A.1

The method given in the solution of Problem 10A.3, based on Eq. 10.4-9 gives the maximum temperature rise in the lubricant as

$$T_{max} - T_1 = \frac{1}{8}\frac{\mu \Omega^2 R^2}{k}$$

$$= \frac{1}{8}\frac{(2.0 g/cm\cdot s)(8000\pi/60 rad/s)^2(2.54cm)^2}{(4\times10^{-4}\times 4.1840\times10^{7}\ g\cdot cm/s^2 K)} = 16.9C$$

Maximum temperature $T_{max} = 217C$

Since T_1 and T_κ are equal, Eq. 11.4-14 is applicable and gives

$$\xi_{max} = \sqrt{\frac{2\ln(1/\kappa)}{((1/\kappa)^2-1)}}$$

$\kappa = 1/1.002$

$$\xi_{max} = \sqrt{\frac{2\ln(1.002)}{(1.002)^2-1}} = 0.99$$

To evaluate T_{max}, we multiply Eq. 11.4-13 by $(T_1 - T_\kappa)$ to make each term finite

$$T - T_\kappa = \frac{\mu \Omega R^2}{k}\cdot\frac{1}{((1/\kappa)^2-1)^2}\left[\left(1-\frac{1}{\xi^2}\right)-\left(1-\frac{1}{\kappa^2}\right)\frac{\ln\xi}{\ln\kappa}\right]$$

with $\xi = \xi_{max}$

$$T_{max} - T_\kappa = 8.438\times10^{6}\cdot[2\times10^{6}] = 16.9\,C$$

which is in agreement with the previous calculation

11.B.2 From energy equation:

a) $$\rho\hat{C}_p v_z\frac{\partial T}{\partial z} = k\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \mu\left(\frac{\partial v_z}{\partial r}\right)^2$$

Velocity profile for a Newtonian fluid in a circular tube: $v_z = v_{z,max}[1-(r/R)^2]$

If we subsitute that into the energy equation we get Eq. 11B.2-1

$$\rho\hat{C}_p\left[1-\left(\frac{r}{R}\right)^2\right]\frac{\partial T}{\partial z} = k\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \frac{4\mu v_{z,max}^2}{R^2}\left(\frac{r}{R}\right)^2$$

b) We Can Simplify this equation

$$\rho \hat{C}_p \left[1-\left(\frac{r}{R}\right)^2\right]\frac{\partial T}{\partial z} = k\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \frac{4\mu v_{z\,max}^2}{R^2}\left(\frac{r}{R}\right)^2$$

using @ large z $\frac{\partial T}{\partial z} = 0$

$$0 = k\frac{1}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right) + \frac{4\mu v_{z,max}^2}{R^2}\left(\frac{r}{R}\right)^2$$

then rearrange

$$\cancel{r}\cdot\frac{1}{\cancel{r}}\frac{d}{dr}\left(r\frac{dT}{dr}\right) = -\frac{4\mu v_{z\,max}^2}{kR^2}\left(\frac{r}{R}\right)^2\cdot r$$

$$\frac{d}{dr}\left(r\frac{dT}{dr}\right) = -\frac{4\mu v_{z\,max}^2}{ke^4}r^3$$

Integrate

$$\int\frac{d}{dr}\left(r\frac{dT}{dr}\right) = \int -\frac{4\mu v_{z\,max}^2}{ke^4}r^3$$

$$\cancel{r}\frac{dT}{dr} = \frac{\mu v_{z\,max}^2}{kR^4}\frac{r^{\cancel{4}\,3}}{r} + \frac{C_1}{r}$$

Integrate again

$$\int\frac{dT}{dr} = \int\frac{\mu v_{z\,max}^2}{kR^4}r^3 + \frac{C_1}{r}$$

$$T = \frac{\mu v_{z,max}^2}{4ke^4}r^4 + C_1\ln r + C_2$$

Evaluate Boundary Conditions

At $r=0$ $\frac{dT}{dr} = 0$ which means $C_1 = 0$

At $T=T_0$ at $r=R$

Which means

$$T_0 = -\frac{\mu V_{z\,max}^2}{4kR^4} R^4 + C_2$$

$$C_2 = T_0 + \frac{\mu V_{z\,max}^2}{4k}$$

Final Temperature Profile

$$T = T_0 + \frac{\mu V_{z\,max}^2}{4k}\left[1 - \left(\frac{r}{R}\right)^4\right]$$

$$T - T_0 = \frac{\mu V_{z\,max}^2}{4k}\left[1 - \left(\frac{r}{R}\right)^4\right]$$

c) $$\rho \hat{C}_p V_{z\,max}\left[1 - \left(\frac{r}{R}\right)^2\right]\frac{\partial T}{\partial z} = k\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \frac{4\mu V_{z\,max}^2}{R^2}\left(\frac{r}{R}\right)^2$$

$$T - T_1 = \left(\frac{4\mu V_{z\,max}}{\rho \hat{C}_p R^2}\right) z + f(r)$$

$$\frac{\partial T}{\partial z} = \frac{4\mu V_{z,max}}{\rho \hat{C}_p R^2}$$

$$\frac{\partial T}{\partial r} = f'(r)$$

Subsitute into the energy equation

$$\cancel{\rho \hat{C}_p} V_{z,max}\left[1 - \left(\frac{r}{R}\right)^2\right]\left(\frac{4\mu V_{z\,max}}{\cancel{\rho \hat{C}_p} R^2}\right) = k\frac{1}{r}\frac{d}{dr}\left(r f'(r)\right) + \frac{4\mu V_{z\,max}^2}{R^2}\left(\frac{r}{R}\right)^2$$

Simplify:

$$✓ \cdot K \cancel{\frac{1}{r}} \frac{d}{dr}(rf') = \frac{4\mu V_{z,max}}{R^2}\left[1 - 2\left(\frac{r}{R}\right)^2\right] \cdot ✓$$

$$\int \frac{d}{dr}(rf') = \int \frac{4\mu V_{z\,max}^2}{KR^2}\left[r - \frac{2r^3}{R^2}\right]$$

$$rf' = \frac{4\mu V_{z\,max}^2}{KR^2}\left(\frac{r^2}{2} - \frac{r^4}{2R^2}\right) + C_1$$

Boundary conditions:

@ $r=0$, $\frac{df}{df}$ finite, therefore

$$C_1 = 0$$

Integrate again:

$$f(r) = \frac{2\mu V_{z\,max}^2}{KR^2}\left(\frac{r^2}{2} - \frac{r^4}{4R^2}\right) + C_2$$

Simplify:

$$f(r) = \frac{\mu V_{z,max}^2}{K}\left[\left(\frac{r}{R}\right)^2 - \frac{1}{2}\left(\frac{r}{R}\right)^4\right] + C_2$$

From earlier

$$C_2 = \frac{-\mu V_{z,max}^2}{4K}$$

$$T - T_1 = \frac{4\mu V_{z\,max}}{\rho c_p R^2} z + \frac{\mu V_{z,max}^2}{K}\left[\left(\frac{r}{R}\right)^2 - \frac{1}{2}\left(\frac{r}{R}\right)^4 - \frac{1}{4}\right]$$

11.6.4

At steady state $\frac{\partial T}{dt} = 0$, and no velocity so left side is 0

For spherical coordinate

$$0 = k\left[\frac{1}{r^2}\cancel{\frac{\partial}{\partial r}}\left(r\cancel{\frac{\partial T}{\partial r}}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial T}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\cancel{\frac{\partial^2 T}{\partial\phi^2}}\right] + \cancel{\mu\Phi_v}$$

0 0 0

Heat flows in the θ direction

So Eq. 6.9-3 simplifies to

$$\frac{1}{r^2\sin\theta}\frac{d}{d\theta}\left(\sin\frac{dT}{d\theta}\right) = 0$$

Integrate

$$\int\frac{d}{d\theta}\left(\sin\theta\frac{dT}{d\theta}\right) = \int 0$$

$$\sin\theta\,\frac{dT}{d\theta} = C_1$$

Integrate again:

$$\int\frac{dT}{d\theta} = \int\frac{C_1}{\sin\theta}$$

$$T = C_1\ln\left|\tan\tfrac{1}{2}\theta\right| + C_2$$

Boundary conditions

$$T(\theta=\theta_1) = T_1$$

$$T(\theta=(\pi-\theta_1)) = T_2$$

$$T_1 = C_1\ln\tan\tfrac{1}{2}\theta_1 + C_2$$

$$T_2 = C_1\ln\tan\tfrac{1}{2}(\pi-\theta_1) + C_2$$

$$T_1 - T_2 = C_1\left[\ln\tan\tfrac{1}{2}\theta_1 - \ln\tan\tfrac{1}{2}(\pi-\theta_1)\right] =$$

$$T - T_2 = C_1\left[\ln\tfrac{1}{2}\tan\theta - \ln\tan\tfrac{1}{2}(\pi-\theta_1)\right]$$

$$\frac{T-T_2}{T_1-T_2} = \frac{\ln\left[\tan\tfrac{1}{2}\theta\,/\,\tan\tfrac{1}{2}(\pi-\theta_1)\right]}{\ln\left[\tan\tfrac{1}{2}\theta_1\,/\,\tan\tfrac{1}{2}(\pi-\theta_1)\right]}$$

12.B.4

a) For a laminar flow down a vertical wall driven by gravity the momentum balance gives

$$\mu \frac{d^2 v_z}{dy^2} = \rho g$$

Integrate:

$$\int \mu \frac{d^2 v_z}{dy^2} = \int \rho g$$

$$\mu \frac{d v_z}{dy} = \rho g y + C_1$$

Integrate again:

$$\int \mu \frac{d v_z}{dy} = \int \rho g y + C_1$$

$$v_z = \frac{\rho g y^2}{2\mu} + C_1 y + C_2$$

Boundary Conditions:

@ $y = 0$ no slip

$v_z = 0$

$C_2 = 0$

@ $y = \delta$ liquid Surface

$$\frac{d v_z}{dy} = 0$$

$$\frac{\rho g}{\mu} y + C_1 = 0$$

$$C_1 = -\frac{\rho g}{\mu} \delta$$

$$v_z = \frac{\rho g}{2\mu} y^2 - \frac{\rho g}{\mu} \delta y$$

Factor:

$$v_z = \frac{\rho g}{2\mu} (2\delta y - y^2)$$

Maximum velocity occurs at the surface $y = \delta$

$$v_{z,max} = \frac{\rho g \delta^2}{2\mu}$$

Subsitute:

$$\boxed{v_z = v_{z\,max} \left[\frac{2y}{\delta} - \frac{y^2}{\delta^2} \right]}$$

Near the wall

$$y \ll \delta \quad \text{so} \quad \left(\frac{y}{\delta}\right)^2 \approx 0$$

Thus: $$v_z = v_{z,max} \frac{2y}{\delta}$$

Subsitue in $v_{z,max} = \dfrac{\rho g \delta^2}{2\mu}$

$$v_z = \frac{\rho g \delta^{\cancel{2}}}{\cancel{2}\mu} \left(\frac{\cancel{2}y}{\cancel{\delta}} \right)$$

$$\boxed{v_z = \frac{\rho g \delta}{\mu} y}$$

b) Steady state $\frac{\partial T}{\partial t} = 0$

$$\rho \hat{C}_p \left(v_z \frac{\partial T}{\partial z} + v_y \frac{\partial T}{\partial y} \right) = k \left(\frac{\partial^2 T}{\partial z^2} + \frac{\partial^2 T}{\partial y^2} \right)$$

$v_y = 0$ $\qquad \frac{\partial^2 T}{\partial z^2} \ll \frac{\partial^2 T}{\partial y^2}$

$v_z = v_z(y)$

$$\boxed{\rho \hat{C}_p v_z \frac{\partial T}{\partial z} = k \frac{\partial^2 T}{\partial y^2}}$$

Subsitute velocity profile from part a

$$v_z = \frac{\rho g \delta}{\mu} y$$

$$\rho \hat{C}_p \left(\frac{\rho g \delta}{\mu} y \right) \frac{\partial T}{\partial z} = k \frac{\partial^2 T}{\partial y^2}$$

Rearrange:

$$y \frac{\partial T}{\partial z} = \frac{\mu k}{\rho^2 C_p g \delta} \frac{\partial^2 T}{\partial y^2}$$

If we define $\beta = \frac{\mu k}{\rho^2 \hat{C}_p g \delta}$

$$\boxed{y \frac{\partial T}{\partial z} = \beta \frac{\partial^2 T}{\partial y^2}}$$

c) Boundary Conditions

Inlet : $T = T_0 \quad z = 0, \ y > 0$

Far from wall : $T \to T_0 \quad y \to \infty$

At the wall : $T = T_i \quad y = 0 \quad z > 0$

d) dimensionless Variable

$$\theta = \frac{T - T_0}{T_i - T_0} \qquad \text{when } T = T_i \quad \theta = 1$$
$$\qquad \text{when } T = T_0 \quad \theta = 0$$

$$\eta = \frac{y}{(3\beta z)^{1/3}}$$

$$T = T_0 + (T_i - T_0)\,\theta(\eta)$$

We use chain rule

- first derive with respect to y

$$\frac{\partial T}{\partial y} = (T_i - T_0)\,\frac{\partial \theta}{d\eta}\,\frac{\partial \eta}{\partial y}$$

Since $\eta = \dfrac{y}{(3\beta z)^{1/3}}$

$$\frac{d\eta}{dy} = \frac{1}{(3\beta z)^{1/3}}$$

$$\frac{\partial T}{\partial y} = (T_i - T_0)\,\frac{1}{(3\beta z)^{1/3}}\,\frac{d\theta}{d\eta}$$

Second derivative with respect to y

differentiate again:

$$\frac{\partial^2 T}{\partial y^2} = (T_1 - T_0)\frac{1}{(3\beta z)^{2/3}}\frac{d^2\theta}{d\eta^2}$$

$$\eta = \frac{y}{(3\beta z)^{1/3}}$$ differentiate with respect to z

$$\frac{d\eta}{dz} = \frac{-1}{3}\frac{y}{(3\beta)^{1/3} z^{4/3}}$$

Chain rule:

$$\frac{\partial T}{\partial z} = T_1 - T_0 \quad \frac{\partial \theta}{\partial \eta}\frac{\partial \eta}{\partial z}$$

We want $y\frac{\partial T}{\partial z} = \beta\frac{\partial^2 T}{\partial y^2}$

$$y\,(T_1 - T_0)\left(\frac{\partial\theta}{\partial\eta}\right)\left(-\frac{1}{3}\frac{y}{(3\beta)^{1/3} z^{4/3}}\right) = \beta\,(T_1 - T_0)\frac{1}{(3\beta z)^{2/3}}\frac{d^2\theta}{d\eta^2}$$

$$-(T_1 - T_0)\frac{1}{3}y^2(3\beta)^{-1/3} z^{-4/3}\frac{d\theta}{d\eta} = \beta\,(T_1 - T_0)\frac{1}{(3\beta z)^{2/3}}\frac{d^2\theta}{d\eta^2}$$

$$y = \eta(3\beta z)^{1/3}$$

$$y^2 = \eta^2(3\beta z)^{2/3}$$

$$-(T_1 - T_0)\frac{1}{3}\eta^2(3\beta)^{2/3}(3\beta)^{+1/3} z^{-4/3}\frac{d\theta}{d\eta} = (T_1 - T_0)\frac{\beta}{(3\beta)^{2/3} z^{2/3}}\frac{d^2\theta}{d\eta^2}$$

$$-\cancel{(T_1-T_0)}\frac{(3\cancel{B})^{1/3}}{3}\eta^2\cancel{z^{-2/3}}\frac{d\theta}{d\eta}=\cancel{(T_1-T_0)}\frac{\cancel{B^{1/3}}}{3^{2/3}}\cancel{z^{-2/3}}\frac{d^2\theta}{d\eta^2}$$

$$-\frac{1}{3}3^{1/3}\eta^2\frac{d\theta}{d\eta}=\frac{1}{3^{2/3}}\frac{d^2\theta}{d\eta^2}$$

$$\frac{d^2\theta}{d\eta^2}+3\eta^2\frac{d\theta}{d\eta}=0$$

e) let

$$p=\frac{d\theta}{d\eta}$$

then $$\frac{dp}{d\eta}+3\eta^2p=0$$

$$\int\frac{dp}{p}=\int-3\eta^2d\eta$$

$$\ln p=-\eta^3+C$$

$$p=C_1e^{-\eta^3}$$

$$\frac{d\theta}{d\eta}=C_1e^{-\eta^3}$$

$$\int d\theta=\int C_1e^{-\eta^3}d\eta$$

$$\theta = C_1 \int e^{-\eta^3} d\eta + C_2$$

Apply boundary conditions

$$\theta = \frac{\int_{\eta}^{\infty} e^{-\bar{\eta}^3} d\bar{\eta}}{\int_0^{\infty} e^{-\bar{\eta}^3} d\eta}$$

Using the Gamma function

$$\int_0^{\infty} e^{-x^3} dx = \frac{1}{3}\Gamma\left(\frac{1}{3}\right)$$

$$\theta = \frac{1}{\Gamma(1/3)} \int_{\eta}^{\infty} e^{-\bar{\eta}^3} d\bar{\eta}$$

f)

$$q = -K\left(\frac{\partial T}{\partial y}\right) \text{ at } y=0$$

Using $T = T_0 + (T_1 - T_0)\theta$

$$q = -K(T_1 - T_0)\left(\frac{\partial \theta}{\partial y}\right)_{y=0}$$

Using the integral solution

$$\left(\frac{d\theta}{d\eta}\right)_{\eta=0} = \frac{1}{\Gamma(1/3)}$$

$$q_{avg} = \frac{3}{2} \frac{k(T_1 - T_0)}{\Gamma(1/3)} (9\beta L)^{-1/3}$$

$$q_{avg}\big|_{y=0} = \frac{3}{2} \frac{(9\beta L)^{-1/3}}{\Gamma(1/3)} (T_1 - T_0)$$

A solid sphere of radius R is initially at a uniform temperature T_0. At time $t = 0$, the temperature of the sphere's surface is suddenly raised to T_1. Assume the density, heat capacity, and thermal conductivity of the solid are constant.

1- Write down the partial differential equation governing the transient temperature profile within the sphere.

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3- Solve the dimensionless partial differential equation using the method of separation of variables. Express the dimensionless temperature profile as an infinite series and evaluate the integration constants.

4- Plot the dimensionless temperature as function of dimensionless radius for $\tau = 0.1$.

5- Modify the provided finite difference code for the spherical coordinates and reproduce the above plot.

1. $$\rho \hat{C}_p \left(\frac{\partial T}{\partial t} + \cancelto{0}{v_r \frac{\partial T}{\partial r}} + \cancelto{0}{\frac{v_\theta}{r} \frac{\partial T}{\partial \theta}} + \cancelto{0}{\frac{v_\phi}{r \sin\theta} \frac{\partial T}{\partial \phi}} \right) =$$

$$k \left[\frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial T}{\partial r} \right) + \cancelto{0}{\frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial T}{\partial \theta} \right)} + \cancelto{0}{\frac{1}{r^2 \sin^2\theta} \frac{\partial^2 T}{\partial \phi^2}} \right]$$

$$+ \mu \Phi_v$$

$$\rho \hat{C}_p \frac{\partial T}{\partial t} = k \left[\frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial T}{\partial r} \right) \right]$$

$$\frac{\partial T}{\partial t} = \frac{k}{\rho \hat{C}_p} \left[\frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial T}{\partial r} \right) \right]$$

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial T}{\partial r} \right) \right)$$

$T(r,t)$ = temperature

r = radial coordinate

t = time

$\alpha = \frac{k}{\rho c_p}$ = thermal diffusivity

$T(r,0) = T_0$

$\frac{\partial T}{\partial r} = 0$ at $r = 0$

$T(R,t) = T_1$

2) $\theta = \frac{T_1 - T}{T_1 - T_0}$

$\eta = \frac{r}{R}$

$\tau = \frac{\alpha t}{R^2}$

$\theta(T_1 - T_0) = T_1 - T$

$T = \underbrace{\cancel{T_1}}_{\text{constant}} - (T_1 - T_0)\theta$

$\frac{\partial T}{\partial t} = -(T_1 - T_0)\frac{\partial \theta}{\partial t}$

$\tau = \frac{\alpha t}{R^2}$

$\frac{\partial}{\partial t} = \frac{\alpha}{R^2}\frac{\partial}{\partial \tau}$

$$r = R\eta$$

$$\frac{\partial}{\partial r} = \frac{1}{R}\frac{\partial}{\partial \eta}$$

$$\frac{\partial T}{\partial t} = -(T_i - T_0)\frac{\partial \theta}{\partial t}$$

replace this with this

$$\frac{\partial T}{\partial t} = -(T_i - T_0)\frac{\alpha}{R^2}\frac{\partial \theta}{\partial \tau}$$

$$\frac{\partial T}{\partial r} = -(T_i - T_0)\frac{1}{R}\frac{\partial \theta}{\partial \eta}$$